

First-Order ODEs (Zill 2.2–2.5)

$\frac{dy}{dx} = g(x)h(y)$, $\int \frac{dy}{h(y)} = \int g(x) dx + C$ – separable, integrate both sides.

$\frac{dy}{dx} + P(x)y = Q(x)$, $\mu(x) = e^{\int P(x) dx}$, $y = \frac{1}{\mu(x)} \left[\int \mu(x)Q(x) dx + C \right]$ – linear, integrating factor μ .

$M dx + N dy = 0$ exact iff $M_y = N_x$; $f(x, y) = \int M dx + g(y) = C$, $f_y = N$ – potential fn. f , $f_x = M$.

$\mu(x) = e^{\int (M_y - N_x)/N dx}$ (x -only), $\mu(y) = e^{\int (N_x - M_y)/M dy}$ (y -only) – restores exactness for non-exact eqns.

$\frac{dy}{dx} = f(y/x)$, sub $y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$ – homogeneous, reduces to separable in v, x .

$\frac{dy}{dx} + P(x)y = Q(x)y^n$, sub $w = y^{1-n} \Rightarrow \frac{dw}{dx} + (1-n)P(x)w = (1-n)Q(x)$ – Bernoulli, linear in w .

Linear Higher-Order ODEs (Zill 3.1–3.6)

$y_2 = y_1(x) \int \frac{e^{-\int P dx}}{[y_1(x)]^2} dx$ – reduction of order, 2nd solution from known y_1 of $y'' + Py' + Qy = 0$.

$ay'' + by' + cy = 0 \Rightarrow am^2 + bm + c = 0$ – homogeneous, const. coeff. (sub $y = e^{mx}$). *Three root cases:*

Distinct real m_1, m_2 : $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$.

Repeated $m_1 = m_2$: $y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}$.

Complex $m = \alpha \pm \beta i$: $y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$.

n th-order, repeated roots: real m (mult. k) $\rightarrow e^{mx}, x e^{mx}, \dots, x^{k-1} e^{mx}$; complex pair (mult. k) $\rightarrow e^{\alpha x} x^j \cos \beta x, e^{\alpha x} x^j \sin \beta x, j = 0, \dots, k-1$.

Undetermined coeff. – trial y_p : const. $\rightarrow A$; poly. deg. $n \rightarrow A_n x^n + \dots + A_0$; $e^{\alpha x} \rightarrow A e^{\alpha x}$; $\cos \beta x / \sin \beta x \rightarrow A \cos \beta x + B \sin \beta x$; product $\rightarrow$ product of forms. Duplicates y_c ? Multiply guess by x (or x^2).

$y_c = c_1 y_1 + c_2 y_2$, $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$, $u_1' = \frac{-y_2 f(x)}{W}$, $u_2' = \frac{y_1 f(x)}{W}$, $y_p = u_1 y_1 + u_2 y_2$ – variation of parameters.

$a x^2 y'' + b x y' + c y = 0$, sub $y = x^m \Rightarrow am(m-1) + bm + c = 0$ – Cauchy-Euler, $x > 0$. Distinct: $y = c_1 x^{m_1} + c_2 x^{m_2}$; repeated: $y = c_1 x^{m_1} + c_2 x^{m_1} \ln x$; complex: $y = x^\alpha [c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)]$.

$y = y_c + y_p$ – general solution; y_c solves homogeneous eqn, y_p is any one particular solution.

Laplace Transforms (Zill 4.1–4.5)

$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$, $t \geq 0$ – converges for s large, f piecewise cont., exp. order.

$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s)$ – linearity.

Laplace transform pairs:

$\mathcal{L}\{1\} = 1/s$, $s > 0$, $\mathcal{L}\{t\} = 1/s^2$, $s > 0$

$\mathcal{L}\{t^n\} = n!/s^{n+1}$, $n = 1, 2, 3, \dots$, $s > 0$

$\mathcal{L}\{e^{at}\} = 1/(s-a)$, $s > a$

$\mathcal{L}\{\sin kt\} = k/(s^2 + k^2)$, $\mathcal{L}\{\cos kt\} = s/(s^2 + k^2)$, $s > 0$

$\mathcal{L}\{\sinh kt\} = k/(s^2 - k^2)$, $\mathcal{L}\{\cosh kt\} = s/(s^2 - k^2)$, $s > |k|$

$\mathcal{L}\{t \sin kt\} = 2ks/(s^2 + k^2)^2$, $s > 0$

$\mathcal{L}\{t \cos kt\} = (s^2 - k^2)/(s^2 + k^2)^2$, $s > 0$

$\mathcal{L}\{e^{at} \sin kt\} = k/[(s-a)^2 + k^2]$, $s > a$

$\mathcal{L}\{e^{at} \cos kt\} = (s-a)/[(s-a)^2 + k^2]$, $s > a$

$\mathcal{L}\{e^{at} t^n\} = n!/(s-a)^{n+1}$, $s > a$

$\mathcal{L}\{\mathcal{U}(t-a)\} = e^{-as}/s$, $s > 0$, $a \geq 0$ – unit step

$\mathcal{L}\{\delta(t-a)\} = e^{-as}$, $a \geq 0$ – Dirac delta

$\mathcal{L}^{-1}\{aF(s) + bG(s)\} = a\mathcal{L}^{-1}\{F(s)\} + b\mathcal{L}^{-1}\{G(s)\}$ – inverse; partial fractions, invert term by term.

$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$; $\mathcal{L}\{f''(t)\} = s^2 F(s) - sf(0) - f'(0)$; $\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - \dots - f^{(n-1)}(0)$ – transform of derivatives.

$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}$ – transform of an integral.

$\mathcal{L}\{e^{at} f(t)\} = F(s-a)$ – 1st translation thm.; shift, $s \rightarrow s-a$.

$\mathcal{U}(t-a) = 0$ ($t < a$), 1 ($t \geq a$); $\mathcal{L}\{f(t-a)\mathcal{U}(t-a)\} = e^{-as}F(s)$ – 2nd translation thm.; else $\mathcal{L}\{g(t)\mathcal{U}(t-a)\} = e^{-as}\mathcal{L}\{g(t+a)\}$.

$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$ – periodic fn., $f(t+T) = f(t)$.

$\int_0^\infty f(t)\delta(t-a) dt = f(a)$; $\mathcal{L}\{\delta(t-a)\} = e^{-as}$, $a \geq 0$; $\mathcal{L}\{\delta(t)\} = 1$ – Dirac delta (unit impulse).

$(f * g)(t) = \int_0^t f(\tau)g(t-\tau) d\tau$, $\mathcal{L}\{f * g\} = F(s)G(s)$ – convolution theorem.

Quick Reference – First-Order ODE Method by Type

Separable ($dy/dx = g(x)h(y)$) – separate, integrate.

Linear ($y' + P(x)y = Q(x)$) – $\mu = e^{\int P dx}$.

Exact ($M dx + N dy = 0$, $M_y = N_x$) – potential f , $f_x = M$, $f_y = N$, $f(x, y) = C$.

Non-exact – integrating factor $\mu(x)$ or $\mu(y)$ to restore $M_y = N_x$.

Homogeneous ($dy/dx = f(y/x)$) – sub $y = vx$.

Bernoulli ($y' + Py = Qy^n$) – sub $w = y^{1-n}$.

Quick Reference – Root-Case Forms Side by Side

Const. coeff. $ay'' + by' + cy = 0$ ($am^2 + bm + c = 0$) vs. Cauchy-Euler $ax^2 y'' + bxy' + cy = 0$ ($am(m-1) + bm + c = 0$, sub $y = x^m$):

Distinct m_1, m_2 – $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$ vs. $y = c_1 x^{m_1} + c_2 x^{m_2}$.

Repeated m_1 – $y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}$ vs. $y = c_1 x^{m_1} + c_2 x^{m_1} \ln x$.

Complex $\alpha \pm \beta i$ – $y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$ vs. $y = x^\alpha [c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)]$.

Quick Reference – Solving an IVP by Laplace Transform

1) Apply $\mathcal{L}\{\cdot\}$: use $\mathcal{L}\{y^{(n)}\} = s^n Y(s) - s^{n-1}y(0) - \dots - y^{(n-1)}(0)$ to get an algebraic eqn. in $Y(s)$.

2) Solve algebraically for $Y(s)$.

3) Partial fractions to match table entries; shifting thms. for $e^{\pm as}$ factors, convolution thm. for a product $F(s)G(s)$.

4) Invert term by term: $y(t) = \mathcal{L}^{-1}\{Y(s)\}$.

Vector Functions & Space Curves (Zill 9.1–9.3)

$\vec{r}(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}$, $\vec{r}'(t) = f'(t)\hat{i} + g'(t)\hat{j} + h'(t)\hat{k}$ – $r'(t)$ tangent to curve where $r'(t) \neq \vec{0}$.

$\vec{T}(t) = \frac{\vec{r}'(t)}{\|\vec{r}'(t)\|}$ – unit tangent vector; unit length, points along increasing t .

$L = \int_a^b \|\vec{r}'(t)\| dt = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2} dt$ – arc length.

$s(t) = \int_a^t \|\vec{r}'(u)\| du$, $\frac{ds}{dt} = \|\vec{r}'(t)\| = v(t)$ – arc length fn. & speed; arc-length reparam. gives unit speed.

$\kappa = \frac{\|\vec{T}'(t)\|}{\|\vec{r}'(t)\|^3} = \frac{\|\vec{r}'(t) \times \vec{r}''(t)\|}{\|\vec{r}'(t)\|^3}$ – curvature; cross-product form usually fastest by hand.

$\vec{N}(t) = \frac{\vec{T}'(t)}{\|\vec{T}'(t)\|}$, $\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$ – principal normal & binormal; T, N, B right-handed orthogonal frame.

$\frac{d\vec{T}}{ds} = \kappa\vec{N}$, $\frac{d\vec{N}}{ds} = -\kappa\vec{T} + \tau\vec{B}$, $\frac{d\vec{B}}{ds} = -\tau\vec{N}$ – Frenet–Serret formulas; τ (torsion) = twisting out of osculating plane.

$\vec{v}(t) = \vec{r}'(t)$, $\vec{a}(t) = \vec{r}''(t) = a_T\vec{T} + a_N\vec{N}$, $a_T = v'(t)$, $a_N = \kappa[v(t)]^2$ – velocity, accel., tangential/normal components.

Scalar/Vector Fields, Gradient, Divergence, Curl (Zill 9.5–9.7)

$\phi(x, y, z)$ – scalar field (temperature, pressure); $\vec{F} = P\hat{i} + Q\hat{j} + R\hat{k}$ – vector field (velocity, force, flux).

$D_{\vec{u}}f = \nabla f \cdot \vec{u}$, $\|\vec{u}\| = 1$ – directional derivative; rate of change of f toward $\vec{u}$.

$\nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$ – gradient; points toward max increase, normal to level surface $f = c$.

$\text{div } \vec{F} = \nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$ – divergence; net outward flux/volume, source if > 0 , sink if < 0 .

$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix}$ – curl; local rotation, $\vec{0}$ everywhere $\Rightarrow$ irrotational.

$\nabla \times (\nabla f) = \vec{0}$; $\nabla \cdot (\nabla \times \vec{F}) = 0$ – div/curl identities; conservative fields are irrotational.

$\nabla^2 f = \nabla \cdot \nabla f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$ – Laplacian; f harmonic if $\nabla^2 f = 0$.

Line Integrals, Path Independence & Conservative Fields (Zill 9.8–9.9)

$\int_C f(x, y, z) ds = \int_a^b f(x(t), y(t), z(t)) \|\vec{r}'(t)\| dt$ – line integral, scalar field (mass of wire, avg. value).

$W = \int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt = \int_C P dx + Q dy + R dz$ – work; reversing orientation of C negates it.

$\vec{F} = \nabla f \Rightarrow \int_C \vec{F} \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$ – fund. thm. for line integrals; endpoints only, path-independent.

Plane $\vec{F} = P\hat{i} + Q\hat{j}$ conservative $\iff \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$; space $\iff \nabla \times \vec{F} = \vec{0}$ – test for a conservative field; requires F cont. diff. on simply connected region.

$f(x, y) = \int P(x, y) dx + g(y)$, match $\frac{\partial f}{\partial y} = Q(x, y)$ to solve $g(y)$ – finding a potential function; same technique as exact DEs.

$\vec{F}$ conservative $\iff \oint_C \vec{F} \cdot d\vec{r} = 0$ for every closed C – closed-loop property.

Green's Theorem, Surface Integrals & Flux (Zill 9.10–9.13)

C positively oriented, piecewise-smooth, simple closed, bounding region D , P, Q cont. 1st partials on open region containing D :

$$\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Green's theorem; C traversed counterclockwise.

$A = \oint_C x dy = -\oint_C y dx = \frac{1}{2} \oint_C (x dy - y dx)$ – area via Green's theorem.

$\vec{r}(u, v) = x(u, v)\hat{i} + y(u, v)\hat{j} + z(u, v)\hat{k}$, $A(S) = \iint_D \|\vec{r}_u \times \vec{r}_v\| dA$ – parametrized surface & area; $\vec{r}_u \times \vec{r}_v$ normal, magnitude scales area.

$\iint_S f(x, y, z) dS = \iint_D f(\vec{r}(u, v)) \|\vec{r}_u \times \vec{r}_v\| dA$ – surface integral, scalar field; for $z = g(x, y)$: $dS = \sqrt{g_x^2 + g_y^2 + 1} dA$.

$\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \hat{n} dS = \iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) dA$ – flux integral; net flow of F through S along $\hat{n}$.

$\iint_S \vec{F} \cdot d\vec{S} = \iint_D \left(-P \frac{\partial g}{\partial x} - Q \frac{\partial g}{\partial y} + R \right) dA$ – flux through $z = g(x, y)$, upward normal; D = projection onto xy -plane, flip sign for downward.

Div. & Stokes' Theorems (Zill 9.14–9.16)

Solid Q bounded by closed, piecewise-smooth S , outward unit normal $\hat{n}$, F cont. 1st partials on Q :

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_Q \nabla \cdot \vec{F} dV$$

Divergence thm. (Gauss); flux (surface) $\rightarrow$ triple (volume) integral.

Oriented, piecewise-smooth S bounded by simple closed piecewise-smooth C (oriented with $\hat{n}$), F cont. 1st partials on open region containing S :

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

Stokes' theorem; closed line integral (circulation) $\rightarrow$ surface integral of curl.

$\hat{n} = \hat{k}$, $S \subset xy$ -plane $\implies \oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{k} dA = \oint_C P dx + Q dy$ – Green's is the flat, 2-D case of Stokes'.

$x = r \cos \theta$, $y = r \sin \theta$, $z = z$, $dV = r dz dr d\theta$ – cylindrical coords; natural for an axis of symmetry.

$x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$, $dV = \rho^2 \sin \phi d\rho d\phi d\theta$ – spherical coords; ρ = dist. from origin, $\phi \in [0, \pi]$ from $+z$ -axis, θ azimuthal.

$\iint_D f(x, y) dA = \iint_D f(r \cos \theta, r \sin \theta) r dr d\theta$ – double integral, polar coords; discs, annuli, sectors.

Quick Reference – Four Fundamental Theorems

Line integrals ($\vec{F} = \nabla f$): $\int_C \nabla f \cdot d\vec{r} = f(\text{end}) - f(\text{start})$.

Green's ($C = \partial D$, plane D): $\oint_C P dx + Q dy = \iint_D (Q_x - P_y) dA$.

Stokes' ($C = \partial S$, oriented S): $\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\text{curl } \vec{F}) \cdot d\vec{S}$.

Divergence ($S = \partial Q$, solid Q): $\iint_S \vec{F} \cdot d\vec{S} = \iiint_Q (\text{div } \vec{F}) dV$.

All four: boundary integral = integral of a derivative-type quantity over the region bounded – same idea as the ordinary FTC, one dimension at a time.

Quick Reference – Which Theorem Applies

$\vec{F} = \nabla f$, want $\int_C \vec{F} \cdot d\vec{r}$ along an open curve $\rightarrow$ fund. thm. for line integrals (endpoints only).

Closed curve C bounding plane region D , want $\oint_C P dx + Q dy \rightarrow$ Green's theorem.

Closed curve C bounding oriented surface S in space, want $\oint_C \vec{F} \cdot d\vec{r} \rightarrow$ Stokes' (reduces to Green's when S is flat).

Closed surface S bounding solid Q , want flux $\iint_S \vec{F} \cdot d\vec{S} \rightarrow$ divergence theorem.

Quick Reference – Space-Curve Quantities at a Glance

$\vec{T} = \vec{r}' / \|\vec{r}'\|$ (unit tangent), $\vec{N} = \vec{T}' / \|\vec{T}'\|$ (principal normal), $\vec{B} = \vec{T} \times \vec{N}$ (binormal) – Frenet frame.

$\kappa = \|\vec{r}' \times \vec{r}''\| / \|\vec{r}'\|^3$ (curvature); τ from Frenet–Serret (torsion).

$L = \int_a^b \|\vec{r}'(t)\| dt$ (arc length); $v(t) = ds/dt = \|\vec{r}'(t)\|$ (speed).

$\vec{a} = a_T \vec{T} + a_N \vec{N}$, $a_T = v'(t)$ (changes speed), $a_N = \kappa v(t)^2$ (changes direction).

Quick Reference – Field Operators at a Glance

∇f – vector; max increase of scalar field f ; normal to level surfaces.

$\nabla \cdot \vec{F}$ – scalar; net outward flux/volume (source if > 0 , sink if < 0).

$\nabla \times \vec{F}$ – vector; rotation/circulation density; $\vec{0}$ everywhere $\Rightarrow$ irrotational.

$\nabla^2 f = \nabla \cdot \nabla f$ – scalar; sum of unmixed 2nd partials; harmonic where $= 0$.

$\nabla \times (\nabla f) = \vec{0}$, $\nabla \cdot (\nabla \times \vec{F}) = 0$ always – gradients irrotational, curls divergence-free.

Quick Reference – Volume Elements by Coordinate

Cartesian: $dA = dx dy$, $dV = dx dy dz$.

Polar: $dA = r dr d\theta$, $x = r \cos \theta$, $y = r \sin \theta$.

Cylindrical: $dV = r dz dr d\theta$, $x = r \cos \theta$, $y = r \sin \theta$, $z = z$.

Spherical: $dV = \rho^2 \sin \phi d\rho d\phi d\theta$, $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$.

Quick Reference – Exact DEs vs. Conservative Fields

Exact DE: $M dx + N dy = 0$ exact $\iff M_y = N_x$; potential f , $f_x = M$, $f_y = N$, $f(x, y) = C$.

Conservative field: plane $\vec{F} = P\hat{i} + Q\hat{j}$ conservative $\iff P_y = Q_x$; space $\iff \nabla \times \vec{F} = \vec{0}$; potential f , $\nabla f = \vec{F}$.

Same integrate-and-match procedure builds the potential – exact DEs and conservative fields are the same idea.